

KevaBook Requirements Specification

1. Introduction

1.1 Project Overview

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for KevaBook, a cloud-based booking platform designed to simplify appointment scheduling and booking management for service-based businesses.

The purpose of this document is to establish a clear understanding of the system's expected functionality, quality attributes, and business objectives. It serves as the primary reference for stakeholders, software developers, testers, and project supervisors throughout the software development lifecycle.

This document focuses on what the system shall do rather than how it will be implemented. Detailed design decisions, architecture, domain modeling, database design, and workflow specifications are documented separately.

1.2 Project Overview

KevaBook is a Software-as-a-Service (SaaS) platform that enables service-based businesses to manage appointments through a centralized online booking system.

The platform allows businesses to create and manage their profiles, register staff members, define the services they offer, configure staff availability, and receive online bookings from clients. Clients can discover businesses, browse available services, select an available staff member, and book appointments without requiring manual communication through phone calls, emails, or messaging applications.

KevaBook follows a multi-tenant architecture in which multiple independent businesses operate within a shared platform while maintaining complete separation of their data. Each Business manages its own staff, services, bookings, locations, and business settings independently.

The platform aims to improve scheduling efficiency, reduce administrative overhead, minimize booking conflicts, and provide a convenient self-service booking experience for both businesses and clients.

1.3 Problem Statement

Many freelancers and small service-based businesses continue to manage appointments manually through phone calls, emails, social media platforms, and messaging applications such as WhatsApp. These approaches often result in scheduling conflicts, double bookings, missed appointments, inconsistent record keeping, and inefficient communication.

In addition, many small businesses lack the financial and technical resources required to develop and maintain dedicated booking systems. Existing enterprise booking solutions are often expensive, overly complex, or designed for larger organizations rather than small and medium-sized service providers.

KevaBook addresses these challenges by providing an affordable, centralized, and easy-to-use booking platform that enables businesses to manage appointments digitally while allowing clients to book services anytime through a self-service interface.

1.4 Project Objectives

The primary objectives of KevaBook are to:

- Provide an online self-service appointment booking platform.
- Enable businesses to manage staff, services, and availability schedules.
- Prevent scheduling conflicts through controlled booking management.
- Simplify appointment management for service providers.
- Improve the overall booking experience for clients.
- Support multiple independent businesses within a single platform through multi-tenancy.
- Deliver timely booking notifications to businesses and clients.
- Provide a scalable foundation capable of supporting future platform enhancements.

1.5 Target Users

KevaBook is intended for the following user groups:

Business Owners

Business owners register and manage their businesses on the platform. They are responsible for managing staff members, services, availability schedules, bookings, business information, and operational settings.

Staff Members

Staff members provide services offered by a Business. Their availability schedules determine when appointments can be booked by clients.

Clients

Clients use the platform to discover businesses, browse available services, select an available staff member, and book appointments without requiring an authenticated account.

1.6 Project Scope

The scope of KevaBook includes:

- Business registration and management.
- Staff management.
- Service catalog management.
- Staff availability management.
- Online appointment booking.
- Booking management.
- Booking confirmation and cancellation.
- Email and SMS notification support.
- Business search and service discovery.
- Multi-tenant data isolation.

The initial release excludes advanced features such as:

- Automatic staff assignment.
- Online payment processing.
- Customer reviews and ratings.
- Loyalty and rewards programs.
- Calendar synchronization with third-party platforms.
- Staff leave management.
- Advanced business analytics and reporting.

These features may be considered for future releases.

1.7 Assumptions

The following assumptions apply to this project:

- Businesses are responsible for maintaining accurate staff, service, and availability information.
- Clients provide accurate contact information during the booking process.
- Internet connectivity is available when accessing the platform.
- Notifications depend on the availability of external email and SMS providers.
- Each Business operates independently within the platform and manages only its own data.

1.8 Constraints

The following constraints apply to the current version of KevaBook:

- Clients may create bookings without registering an account.
- Each booking is associated with a single Business, a single Service, and a single Staff member.
- A Staff member cannot be assigned overlapping bookings.
- Bookings must occur within the configured availability schedule of the assigned Staff member.
- The system supports only the features defined within the Minimum Viable Product (MVP) scope.

2. Functional Requirements

This section defines the functional capabilities that the KevaBook platform shall provide. Each requirement describes a specific system behavior that must be implemented to satisfy the business objectives of the platform.

2.1 User Management

FR-01 User Registration

The system shall allow a Business Owner to register an account on the platform.

FR-02 User Authentication

The system shall authenticate Business Owners before granting access to protected system resources.

FR-03 User Profile Management

The system shall allow authenticated Business Owners to view and update their profile information.

2.2 Business Management

FR-04 Business Creation

The system shall allow an authenticated Business Owner to create a Business profile.

FR-05 Business Profile Management

The system shall allow Business Owners to update their Business information, including business name, description, contact information, and operating status.

FR-06 Business Search

The system shall allow Clients to search for Businesses using supported search criteria.

2.3 Staff Management

FR-07 Staff Creation

The system shall allow Business Owners to create Staff members associated with their Business.

FR-08 Staff Management

The system shall allow Business Owners to update Staff information.

FR-09 Staff Activation

The system shall allow Business Owners to activate or deactivate Staff members.

FR-10 Staff Listing

The system shall allow Business Owners to view all Staff members belonging to their Business.

2.4 Service Management

FR-11 Service Creation

The system shall allow Business Owners to create Services offered by their Business.

FR-12 Service Management

The system shall allow Business Owners to update service information, including name, description, duration, pricing, and availability status.

FR-13 Service Deactivation

The system shall allow Business Owners to activate or deactivate Services.

FR-14 Service Listing

The system shall allow Business Owners to view all Services offered by their Business.

2.5 Staff Availability Management

FR-15 Availability Configuration

The system shall allow Business Owners to define and manage the availability schedules of individual Staff members.

FR-16 Availability Update

The system shall allow Business Owners to modify Staff availability schedules.

FR-17 Availability Retrieval

The system shall display available booking time slots based on the configured availability of the selected Staff member.

2.6 Booking Management

FR-18 Booking Creation

The system shall allow Clients to create bookings for available Services.

FR-19 Staff Selection

The system shall allow Clients to select an available Staff member when creating a booking.

FR-20 Booking Validation

The system shall validate booking requests against Staff availability and existing bookings before confirming the reservation.

FR-21 Booking Confirmation

The system shall confirm successful bookings and record the reservation.

FR-22 Booking Modification

The system shall allow Business Owners to update booking information when necessary.

FR-23 Booking Cancellation

The system shall allow Clients and Business Owners to cancel bookings in accordance with the platform's booking policies.

FR-24 Booking History

The system shall allow Business Owners to view current and historical bookings associated with their Business.

2.7 Search and Discovery

FR-25 Business Discovery

The system shall allow Clients to browse available Businesses.

FR-26 Service Discovery

The system shall allow Clients to browse Services offered by a selected Business.

FR-27 Staff Availability Search

The system shall display available appointment slots for the selected Service and Staff member.

2.8 Notification Management

FR-28 Booking Confirmation Notification

The system shall send booking confirmation notifications to Clients and Business Owners after a successful booking.

FR-29 Booking Cancellation Notification

The system shall notify affected parties when a booking is cancelled.

FR-30 Booking Reminder Notification

The system shall send reminder notifications prior to scheduled appointments.

2.9 Multi-Tenant Management

FR-31 Tenant Isolation

The system shall ensure that each Business can access and manage only its own data.

FR-32 Data Ownership

The system shall associate all operational records with the Business that owns them.

3. Non-Functional Requirements

This section defines the quality attributes and operational characteristics that the KevaBook platform shall satisfy. These requirements ensure that the system is secure, reliable, scalable, maintainable, and capable of providing a consistent user experience.

3.1 Performance

NFR-01 Response Time

The system shall respond to user requests within an acceptable time under normal operating conditions.

NFR-02 Concurrent Users

The system shall support multiple concurrent users performing booking and management operations without significant degradation in performance.

NFR-03 Search Performance

The system shall efficiently retrieve Businesses, Services, and available booking slots.

3.2 Scalability

NFR-04 Business Growth

The system shall support the addition of new Businesses without affecting the operation of existing tenants.

NFR-05 Horizontal Scalability

The platform shall support scaling to accommodate increasing numbers of users, bookings, and businesses.

3.3 Availability

NFR-06 System Availability

The platform shall provide a high level of availability to ensure that booking services remain accessible to users.

NFR-07 Fault Tolerance

The system shall continue operating where possible despite the failure of individual components or services.

3.4 Security

NFR-08 Authentication

The system shall require authenticated access to protected business management functions.

NFR-09 Authorization

The system shall ensure that users can access only the resources and operations permitted by their assigned roles.

NFR-10 Data Protection

The system shall protect sensitive business and client information from unauthorized access.

3.5 Multi-Tenant Isolation

NFR-11 Tenant Data Isolation

The platform shall ensure complete logical isolation of data between Businesses.

NFR-12 Tenant Data Ownership

All operational data shall remain associated with the Business that owns it throughout its lifecycle.

3.6 Reliability

NFR-13 Data Consistency

The system shall maintain consistent booking data during concurrent operations.

NFR-14 Event Reliability

The platform shall reliably process business events such as booking creation, confirmation, cancellation, and notifications.

3.7 Maintainability

NFR-15 Modular Design

The system shall be designed to support future enhancements with minimal impact on existing functionality.

NFR-16 Extensibility

The platform shall support the addition of new features and business capabilities without requiring major architectural changes.

3.8 Usability

NFR-17 Ease of Use

The platform shall provide an intuitive and user-friendly interface for both Business Owners and Clients.

NFR-18 Responsive User Interface

The platform shall provide a consistent user experience across desktop, tablet, and mobile devices.

3.9 Auditability

NFR-19 Audit Information

The system shall maintain audit information for significant business operations, including entity creation and modification.

3.10 Time Management

NFR-20 Time Zone Support

The platform shall support Business-specific time zones to ensure accurate scheduling and booking operations.

3.11 Data Integrity

NFR-21 Booking Integrity

The system shall prevent inconsistent booking data, including duplicate or overlapping bookings for the same Staff member.

NFR-22 Referential Integrity

The system shall maintain valid relationships between Businesses, Staff, Services, Availability schedules, Clients, and Bookings.

4. Use Case Summary

This section provides a high-level summary of the primary interactions between users and the KevaBook platform. It identifies the system actors and the major use cases they perform.

4.1 System Actors

The KevaBook platform involves the following primary actors:

Business Owner

A Business Owner is responsible for managing business operations within the platform. This includes creating and maintaining business information, managing staff, configuring services and availability, and monitoring bookings.

Staff Member

A Staff Member provides services offered by a Business. Staff members operate according to the availability schedules configured for them and fulfil client bookings.

Client

A Client is an external user who searches for businesses, browses available services, and creates bookings through the platform. Clients are not required to register before making a booking.

4.2 Business Owner Use Cases

Use case ID	Use case	Description
UC-01	Register Account	Create a business owner account
UC-02	Sign in	Authenticate and access platform
UC-03	Manage business	Create and update business information
UC-04	Manage staff	Create, update, activate, deactivate staff members
UC-05	Manage service	Create, update, activate, deactivate services
UC-06	Manage staff availability	Configure,update staff working schedules
UC-07	Manage bookings	View, update and cancel booking
UC-08	View booking history	View current and historical bookings

4.3 Client Use Cases

Use case ID	Use case	Description
UC-09	Search businesses	Search and browse business on the platform
UC-10	View services	Browse services offered by a business
UC-11	View staff availability	View available booking slots for a selected staff member
UC-12	Create booking	Book a service with an available staff
UC-13	Cancel booking	Cancel an existing booking
UC-14	Receive notification	Receive booking confirmations, reminders and cancellation notifications

4.4 Staff Member Use Cases

The initial release of KevaBook does not provide a dedicated Staff portal. Staff members participate in the booking process by delivering services according to their assigned schedules.

Future versions of the platform may introduce Staff-specific functionality, including:

- Viewing personal schedules
- Managing availability
- Viewing assigned bookings
- Receiving booking notifications

5. References to Supporting Documents

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the KevaBook platform. Detailed design and implementation information is documented separately to maintain a clear separation between requirements and system design.

The following documents complement this specification:

Document	Description
KevaBook Domain Model	It defines the core business entities, their attributes, relationships and the logical structure of the booking domain
KevaBook Business Rules	It defines the business policies, constraints, and validation rules governing the booking operations, staff scheduling and tenant isolation
KevaBook System Architecture	It describes the overall system architecture, microservices, service responsibilities, communication patterns, security architecture, and deployment approach.
KevaBook Database Design	It documents the database strategy, schemas, entity relationships, and persistence design for each microservice.
KevaBook Workflow Design	It illustrates the business processes and system interactions using workflow and sequence diagrams for key platform operations.
KevaBook API Specification	It defines the REST APIs, request and response models, error handling, authentication requirements and event contracts between services.

These documents should be read together to obtain a complete understanding of the KevaBook platform, from business requirements through system architecture and implementation.